12. a. Write a C program to Encrypt the message “meet me at the usual place at ten rather than eight oclock” using the Hill cipher with the key.

9 4

5 7

a. Show your calculations and the result.

b. Show the calculations for the corresponding decryption of the ciphertext to recover the original plaintext.

Aim:

To encrypt the message “meet me at the usual place at ten rather than eight oclock” using the Hill cipher with the given key matrix, and decrypt it to recover the original plaintext.

Procedure:

1. Read the plaintext and remove spaces.
2. Convert letters into numbers from 0 to 25.
3. Divide the plaintext into pairs of letters.
4. Multiply each pair by the key matrix modulo 26.
5. Display the ciphertext.
6. Use the inverse key matrix to decrypt the ciphertext**.**

Code:

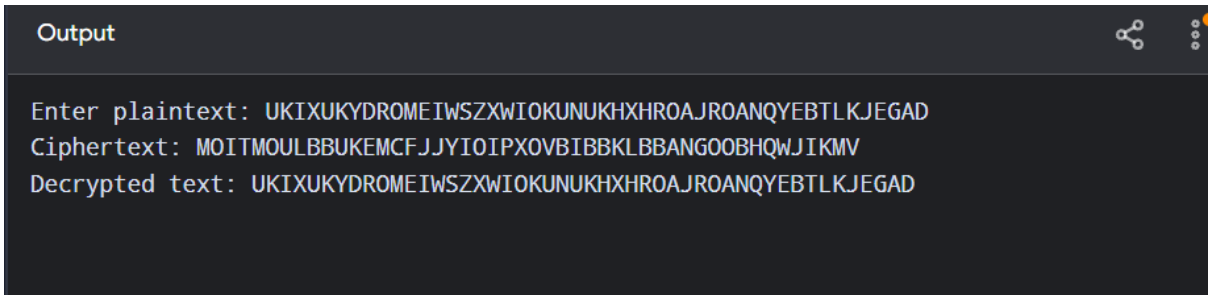
```
#include <stdio.h>
#include <ctype.h>
#include <string.h>
int main() {
   char text[200], cipher[200], plain[200];
   int i, n = 0, a, b, x, y;
   int k[2][2] = {{9, 4}, {5, 7}};
   int inv[2][2] = {{5, 12}, {15, 25}};
   printf("Enter plaintext: ");
```

```
    fgets(text, sizeof(text), stdin);
    for (i = 0; text[i]; i++)
        if (isalpha((unsigned char)text[i]))
            text[n++] = toupper(text[i]);
    if (n % 2 != 0)
      text[n++] = 'X';
    text[n] = '\0';
    for (i = 0; i < n; i += 2) {
        a = text[i] - 'A';
        b = text[i + 1] - 'A';
        cipher[i] = (k[0][0]*a + k[0][1]*b) % 26 + 'A';
        cipher[i+1] = (k[1][0]*a + k[1][1]*b) % 26 + 'A';
    }
    cipher[n] = '\0';
    for (i = 0; i < n; i += 2) {
        a = cipher[i] - 'A';
        b = cipher[i+1] - 'A';
        x = (inv[0][0]*a + inv[0][1]*b) % 26;
        y = (inv[1][0]*a + inv[1][1]*b) % 26;
        plain[i] = x + 'A';
        plain[i+1] = y + 'A';
    }
    plain[n] = '\0';
    printf("Ciphertext: %s\n", cipher);
    printf("Decrypted text: %s\n", plain);
    return 0;
}
```

Input:

meet me at the usual place at ten rather than eight oclock

Output:



Result:

Thus, the Hill cipher program was executed successfully. The plaintext was encrypted using the given key matrix and decrypted to recover the original message**.**